

# Betwixt Minimum Variance and Hierarchical Risk Parity

## Analytical Results for the Schur Bridge

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### Abstract

Schur-complementary allocation interpolates hierarchical risk parity ( $\gamma = 0$ ) and minimum variance ( $\gamma \rightarrow 1$ ) through a coupling  $\gamma$  that sets how much of the estimated cross-block covariance the recursion uses. We ask where on the bridge to sit when the covariance is estimated with error, and answer in three layers. A structural theorem shows that top-level cross-block noise is invisible at the hierarchical end and prices coupling at order  $\gamma\tau^2$ , so small noise moves the optimum uniformly away from hierarchical risk parity whenever coupling has first-order population value. A local theorem at an exact minimum-variance endpoint shows that positive marginal noise sensitivity pushes the optimum into the interior. An exchangeable four-asset family is then solved exactly: its unique optimizer is interior at every admissible noise level and decreases monotonically from full coupling to the middle of the bridge. Exact examples mark the boundaries of what holds in general, and every exact claim is verified in rational arithmetic.

**Keywords:** portfolio construction; hierarchical risk parity; Schur complement; estimation error; shrinkage; bias–variance trade-off.

**JEL:** G11, C13, C58. **MSC2020:** 91G10, 62H12.

## 1 Introduction

Hierarchical risk parity [1] allocates by recursive bisection without inverting a covariance matrix. Minimum variance inverts it in full. The two are ends of a single construction: augment each block with a  $\gamma$ -scaled Schur complement of its sibling and the recursion is exactly HRP at  $\gamma = 0$ , while recovering minimum variance as  $\gamma \rightarrow 1$ . This unification first appeared in a blog [2], then in a textbook [3, §12.3.4], with a fuller development in [4]. It quickly found its way into practice: `skfolio`, a popular Python portfolio library, ships it as `SchurComplementary`. On the hierarchical graph structures of [6] the  $\gamma = 1$  recovery is exact. The endpoints have since been compared analytically: [5] derive the weight noise of HRP and of Markowitz under covariance estimation error and prove HRP is the more robust.

The question behind the bridge is a form of the classical estimation-risk problem. [9] quantified the sampling error of efficient portfolios; [10] named the enigma; [11] showed that imposing the wrong constraints acts as shrinkage and helps; [12] documented how hard the  $1/N$  portfolio is to beat; and [13] made shrinkage of the input systematic. The Schur coupling gives that trade-off a single parameter, and the present paper gives the parameter provable structure.

This paper asks where on the bridge one should sit when the covariance is estimated with error, and answers in three layers. (i) *The HRP end, in general.* Top-level cross-block noise is invisible

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at  $\gamma = 0$ , and its expected effect factors exactly as  $F - V_0 = \gamma\tau^2 H$  with  $H$  bounded (Theorem 2). Whenever coupling has first-order population value, the minimizer is uniformly separated from HRP for all sufficiently small noise (Theorem 1). The order  $\gamma\tau^2$  cannot be improved to  $\gamma^2\tau^2$ : an exact example has boundary derivative  $-\frac{1}{700} + \frac{8}{189}\tau^2$ , so HRP becomes a one-sided local minimum at the finite noise level  $\tau^2 = \frac{27}{800}$  (Example 1). (ii) *The minimum-variance end, locally.* If the bridge reaches minimum variance exactly at  $\gamma = 1$ , then  $V'_0(1) = 0$  automatically and  $V''_0(1) \geq 0$  (Lemma 2); if  $V''_0(1) > 0$  and the marginal noise sensitivity  $G'(1)$  is positive, every sufficiently small noise level pushes the optimizer into the interior, by  $1 - \gamma^*(\tau) \sim \tau^2 G'(1)/V''_0(1)$  (Theorem 3). The sign condition is essential. (iii) *An exactly solved family.* For an exchangeable four-asset family with common cross-block noise, the expected out-of-sample variance is computed in closed form; it is strictly convex on  $[0, 1]$ , its unique minimizer  $\gamma^*(e)$  is interior for every admissible noise amplitude  $e \in (0, c]$ , and it decreases monotonically from  $\gamma^*(0) = 1$  to  $\gamma^*(c) = \frac{1}{2}$  (Theorem 4). This is a statement about the exact objective, not a surrogate. One care is required: at large noise the raw recursion has a removable singularity at one interior coupling, so the objective is defined by continuous extension, and the minimizer is proved to sit strictly inside the region where the recursion is defined. A companion exact example with noise in a single covariance entry (Example 2) shows that interior optimality survives without the common-noise symmetry. A clipped monotone frontier for the second-order surrogate (Theorem 5) organizes the general case. We are explicit about what is not claimed: covariance error does not always force an interior optimum, the noise cost is not  $O(\gamma^2\tau^2)$  in general, and either endpoint can be optimal on a nontrivial interval of noise levels.

## 2 The recursion

*The family.* Following [4], fix an asset order and a balanced bisection tree; assume for notational simplicity that  $n$  is a power of two, so sibling blocks have equal size (a step-up matrix enters in the general case [4]). A hierarchical allocation assigns a node's capital to its children ( $L, R$ ) in the ratio  $1/\nu(\cdot_L) : 1/\nu(\cdot_R)$  for an inverse-fitness functional  $\nu$ , and recurses. Writing a node's block of the covariance as  $\begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$ , hierarchical risk parity passes  $A$  and  $D$  to the children unchanged. The Schur-complementary family instead passes *augmented* blocks built from the  $\gamma$ -damped Schur complement and its companion vector,

$$A^c(\gamma) = A - \gamma BD^{-1}B^\top, \quad b_A(\gamma) = \mathbf{1} - \gamma BD^{-1}\mathbf{1}, \quad (1)$$

and symmetrically for  $D^c(\gamma), b_D(\gamma)$ . In the canonical presentation the intra-group matrix is  $A^c(\gamma)$  with an elementwise division by  $b_A b_A^\top$  and the inter-group fitness is evaluated on  $(A^c(\gamma)^{-1} \circ b_A b_A^\top)^{-1}$ . At  $\gamma = 0$  every member is HRP. At  $\gamma = 1$  the scheme replicates the minimum-variance portfolio; the one-step form of that equivalence is Proposition 1, following Appendix A of [4].

**Proposition 1** ( $\gamma = 1$  replicates minimum variance). *Let  $\Sigma \succ 0$  have blocks  $A, B, D$  as above and write  $A^c = A - BD^{-1}B^\top$ ,  $b_A = \mathbf{1} - BD^{-1}\mathbf{1}$ , and symmetrically  $D^c, b_D$ . Then*

$$\Sigma^{-1}\mathbf{1} = \begin{pmatrix} (A^c)^{-1}b_A \\ (D^c)^{-1}b_D \end{pmatrix}. \quad (2)$$

*For any SPD  $Q$  and  $b \neq 0$ , the constrained problem  $w(Q, b) = \arg \min_{w: b^\top w=1} w^\top Q w$  has solution and attained variance*

$$w(Q, b) = \frac{Q^{-1}b}{b^\top Q^{-1}b}, \quad \nu(Q, b) = w(Q, b)^\top Q w(Q, b) = \frac{1}{b^\top Q^{-1}b}, \quad (3)$$

so that  $Q^{-1}b = w(Q, b)/\nu(Q, b)$ . Consequently the two-block scheme that allocates between blocks in the ratio  $1/\nu(A^c, b_A) : 1/\nu(D^c, b_D)$  and within blocks by  $w(A^c, b_A)$ ,  $w(D^c, b_D)$  reproduces  $w \propto \Sigma^{-1}\mathbf{1}$ , and recursive application to the sub-problems (each again of the form  $Q^{-1}b$ ) extends this down the tree. If a block has  $b_A = 0$  then by (2) that block of  $\Sigma^{-1}\mathbf{1}$  is exactly zero and the block receives nothing; the normalized  $w(Q, b)$  is simply not formed. (For instance  $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  has  $b_A = 0$  and  $\Sigma^{-1}\mathbf{1} = (0, 1)^\top$ .)

*Proof.* For (2), multiply the block inversion identity

$$\begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}^{-1} = \begin{pmatrix} (A^c)^{-1} & 0 \\ 0 & (D^c)^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ -B^\top A^{-1} & I \end{pmatrix}$$

on the right by  $\mathbf{1}$ ; the top block is  $(A^c)^{-1}(\mathbf{1} - BD^{-1}\mathbf{1}) = (A^c)^{-1}b_A$  and the bottom block is analogous. For (3), the Lagrangian condition is  $Qw = \lambda b$ , so  $w = \lambda Q^{-1}b$ ; the constraint  $b^\top w = 1$  forces  $\lambda = 1/(b^\top Q^{-1}b)$ , giving the stated  $w(Q, b)$ , and substituting back gives  $\nu(Q, b) = \lambda$ . Rearranging,  $Q^{-1}b = w(Q, b)(b^\top Q^{-1}b) = w(Q, b)/\nu(Q, b)$ . Apply this to each block of (2): the unnormalized block solutions are the normalized sub-portfolios scaled by exactly the inverse fitness  $1/\nu$ , which is the stated two-block scheme.  $\square$

*Remark 1* (Minimum variance in disguise). The constrained problem (3) was stated for general  $b$ , and that generality is not idle. Several portfolio problems that look different are minimum variance with a prescribed unit exposure, minimizing  $w^\top Qw$  subject to  $b^\top w = 1$ , and are therefore minimum variance in disguise: the tangency portfolio ( $b = \mu$ ), maximum diversification ( $b = \sigma$ , the vector of volatilities), and the variance-optimal combination of forecasts, where  $Q$  is the covariance of forecast errors and  $b = \mathbf{1}$  [7]. Tracking error against a benchmark becomes minimum variance in active weights, and hedging a single beta exposure takes  $b$  to be that exposure; their generic versions carry several affine constraints at once and sit outside (3). The block identity (2) and the damped bridge apply to each with  $\mathbf{1}$  replaced by  $b$  throughout. The noise calculus of this paper is developed for  $b = \mathbf{1}$ ; we expect, but do not prove here, that the endpoint results carry over with the companion vectors  $b_A(\gamma)$  built from  $b$ .

*The variant analyzed.* Appendix A of [4] licenses an equivalent “collapse” of the pair  $(A^c, b_A)$  into a single matrix  $\tilde{A}$  with  $\tilde{A}^{-1}\mathbf{1} = A^c(\gamma)^{-1}b_A(\gamma)$ . The reference implementations take exactly this route.<sup>1</sup> It is the recursion we analyze. Each child block is replaced by

$$A_\gamma = \text{sym}\left[(I - \gamma BD^{-1})^{-1}A^c(\gamma)\right], \quad D_\gamma = (A \leftrightarrow D, B \rightarrow B^\top), \quad (4)$$

where  $\text{sym}(X) = \frac{1}{2}(X + X^\top)$ ; note  $(I - \gamma BD^{-1})\mathbf{1} = b_A(\gamma)$ , so (4) is the collapse of (1), up to the symmetrization. The fitness is the *naïve portfolio variance*

$$\nu(X) = \tilde{w}^\top X \tilde{w}, \quad \tilde{w}_i = \frac{1/X_{ii}}{\sum_j 1/X_{jj}}, \quad (5)$$

so the left child receives the fraction  $\nu(D_\gamma)/(\nu(A_\gamma) + \nu(D_\gamma))$  of the node’s capital, and the recursion continues inside each augmented block down to singletons. Nodes with a singleton child perform no augmentation; the implementations skip the augmentation whenever *either* child has one asset, so a two-asset node splits by the inverse of its already augmented diagonal entries. At  $\gamma = 0$  the

<sup>1</sup>`skfolio`’s `SchurComplementary` and the `allocation` package use identical augmentation and fitness routines; see [github.com/skfolio/skfolio](https://github.com/skfolio/skfolio) and [github.com/microprediction/allocation](https://github.com/microprediction/allocation).

augmentation (4) is the identity and the recursion is exactly HRP over the fixed order. Implementations optionally cap  $\gamma$  at an in-sample variance turning point (`keep_monotonic`); we analyze the raw recursion. One thing cannot be simplified away. The inter-group fitness must be a quadratic form evaluated on the *augmented* block. A diagonal-only  $\nu$ , as used in some HRP variants, generally does not preserve the Schur block-inversion identity (2) that exact minimum-variance recovery requires.

Two structural facts hold across the entire family and not just the variant (4): by (1) the cross-block  $B$  enters every member only through the products  $\gamma BD^{-1}(\cdot)$ , and at  $\gamma = 0$  the augmentation is the identity with  $b_A = \mathbf{1}$ .

*The noise model.* Let  $\Sigma \succ 0$  be the population covariance and let the estimate be

$$\hat{\Sigma}_\tau = \Sigma + \tau N, \quad (6)$$

where  $N$  is a random symmetric matrix. The objects of study are the expected out-of-sample variance and its population limit,

$$F(\gamma; \tau) = \mathbb{E}[w(\gamma; \hat{\Sigma}_\tau)^\top \Sigma w(\gamma; \hat{\Sigma}_\tau)], \quad V_0(\gamma) = F(\gamma; 0), \quad (7)$$

where  $w(\gamma; M)$  denotes the allocation the recursion produces from an input matrix  $M$ . The allocation  $w$  and the realized variance  $w^\top \Sigma w$  are rational functions of  $(\gamma, M)$ , hence  $C^\infty$  wherever their denominators do not vanish. When  $N$  has finite support (as in all examples below)  $F$  is itself a rational function of  $\gamma$ ; for general  $N$  it is merely smooth, under the assumptions we now state.

**Assumption 1.** *There are  $\bar{\gamma} \in (0, 1]$ ,  $\bar{\tau} > 0$ , and a compact set  $K = [0, \bar{\gamma}] \times \mathcal{M}$ , with  $\mathcal{M}$  a compact neighborhood of  $\Sigma$  in the symmetric matrices, such that every denominator appearing in the recursion (the inverses in (4), the diagonal entries in (5), and the fitness sums) is bounded away from zero on  $K$ ; and  $N$  is bounded with  $\Sigma + \tau N \in \mathcal{M}$  for all  $|\tau| \leq \bar{\tau}$ .*

**Assumption 2.**  *$N$  is supported on the top-level cross-block ( $N_{LL} = 0$ ,  $N_{RR} = 0$ ) and its law is symmetric:  $N \stackrel{d}{=} -N$ .*

Under Assumption 1,  $(\gamma, \tau) \mapsto w(\gamma; \Sigma + \tau N)$  is  $C^\infty$  on  $[0, \bar{\gamma}] \times [-\bar{\tau}, \bar{\tau}]$  for every realization, with all derivatives bounded uniformly over realizations;  $F$  is then  $C^\infty$  by differentiation under the expectation (the derivatives are dominated by their uniform bounds). Assumption 2 makes  $F$  even in  $\tau$ .

*Remark 2* (Model scope). The model (6)–(7) with Assumption 2 is stylized on purpose: it isolates the cross-block channel, which is the only channel the coupling  $\gamma$  touches. A natural generative alternative is partial information about  $\Sigma$  itself, for instance  $\hat{\Sigma}$  equal to the sample covariance of  $T$  independent draws from  $\mathcal{N}(0, \Sigma)$ , which corresponds to  $\tau \asymp T^{-1/2}$  with noise supported on *every* block. Under whole-matrix noise the exact invariance of Lemma 1 applies only to the cross-block component of the noise:  $F(0; \tau)$  is then itself noisy, and the endpoint comparison becomes a comparison of noise costs rather than a comparison against a noise-free anchor. The analysis of that model is not attempted here; the results below are exact for (6) under Assumptions 1–2 and are offered as the cross-block part of the larger question.

### 3 The hierarchical end

**Lemma 1** (The HRP end is blind to the cross-block). *For a fixed tree,  $w(0; M)$  is a function of the diagonal blocks  $(A, D)$  alone. Consequently, under Assumption 2,  $w(0; \hat{\Sigma}_\tau) = w(0; \Sigma)$  almost surely and  $F(0; \tau) = V_0(0)$  exactly, for every  $\tau$ .*

*Proof.* At  $\gamma = 0$  the augmentation (4) returns its first argument. The top-level split therefore uses  $\nu(A)$  and  $\nu(D)$ , and (5) reads only entries of  $A$  and of  $D$ . The recursion then continues inside the unmodified blocks  $A$  and  $D$ ; every quantity computed below the top level is a function of entries of  $A$  or entries of  $D$ . The top-level cross-block  $B$  appears nowhere. (Cross-blocks interior to  $A$  or  $D$  are read, through the naive variances of their parents; only the top-level  $B$  is never read.) The second sentence follows because  $\hat{\Sigma}_\tau$  and  $\Sigma$  share their diagonal blocks under Assumption 2.  $\square$

*Remark 3* (Scope). The lemma concerns a fixed tree and this particular variant: the recursion reads the naive variance of full child sub-blocks, as in [1]. A split on diagonal variances alone would read nothing off-diagonal. When the tree itself is data-driven (a dendrogram, or Fiedler seriation [4]), the ordering step does read the cross-block, but only through the discrete tree choice, which for a generic estimate is locally constant; the differential statements below are unaffected for noise small enough to leave the tree unchanged. (In Example 2 the seriation returns the same top partition on both noise branches, so fixing the order there is without loss.)

**Theorem 1** (Uniform separation from HRP under small noise). *Suppose Assumption 1 holds,  $N$  is bounded (no support or symmetry restriction needed), and  $V'_0(0) < 0$ , i.e. coupling has first-order population value. Then there exist  $\varepsilon > 0$ ,  $\tau_0 > 0$ , and  $c > 0$  such that*

$$\partial_\gamma F(\gamma; \tau) \leq -c \quad \text{for all } 0 \leq \gamma \leq \varepsilon, \ 0 \leq \tau \leq \tau_0.$$

Consequently every minimizer of  $F(\cdot; \tau)$  over  $[0, \bar{\gamma}]$  lies in  $[\varepsilon, \bar{\gamma}]$ , for every  $\tau \leq \tau_0$ .

*Proof.* Write  $V(\gamma; M) = w(\gamma; M)^\top \Sigma w(\gamma; M)$ . Under Assumption 1,  $\partial_\gamma V$  exists, is continuous, and is bounded on  $K$ , so  $\partial_\gamma F(\gamma; \tau) = \mathbb{E}[\partial_\gamma V(\gamma; \Sigma + \tau N)]$  by differentiation under the expectation, and  $(\gamma, \tau) \mapsto \partial_\gamma F(\gamma; \tau)$  is jointly continuous on  $[0, \bar{\gamma}] \times [0, \bar{\tau}]$  by dominated convergence. Its value at  $(0, 0)$  is  $V'_0(0) < 0$ . By continuity there is a relatively open neighborhood of  $(0, 0)$  on which  $\partial_\gamma F \leq \frac{1}{2}V'_0(0) =: -c < 0$ ; choose  $\varepsilon, \tau_0$  so that  $[0, \varepsilon] \times [0, \tau_0]$  sits inside it. For  $\tau \leq \tau_0$  and any  $\gamma \in [0, \varepsilon]$ , integrating the derivative gives  $F(\varepsilon; \tau) \leq F(\gamma; \tau) - c(\varepsilon - \gamma) < F(\gamma; \tau)$ , so no point of  $[0, \varepsilon]$  can be a minimizer.  $\square$

### 3.1 The size of the noise cost

How large is  $F - V_0$  near the HRP end? The correct answer is order  $\gamma\tau^2$ , and it is worth displaying the mechanism, because a plausible shortcut gives the wrong exponent. Write the estimated cross-block as  $\hat{B} = B + \tau E$  for a realization  $E$  of  $N$ 's cross-block. Expanding (4) at small  $\gamma$ ,

$$A_\gamma(\hat{B}) = A + \gamma \operatorname{sym}(\hat{B}D^{-1}A - \hat{B}D^{-1}\hat{B}^\top) + O(\gamma^2). \quad (8)$$

Averaging over the symmetric branches  $\hat{B} = B \pm \tau E$ , the terms linear in  $\tau$  cancel, but the term quadratic in  $\hat{B}$  does not:

$$\frac{1}{2} \sum_{\pm} A_\gamma(B \pm \tau E) - A_\gamma(B) = -\gamma \tau^2 E D^{-1} E^\top + O(\gamma^2 \tau^2). \quad (9)$$

The expected augmented block is displaced at order  $\gamma\tau^2$ , and the allocation inherits the displacement unless the weight functional happens to annihilate it. In terms of derivatives of the realized variance,

$$\frac{\partial^2}{\partial \tau^2} (w^\top \Sigma w) = 2 w_\tau^\top \Sigma w_\tau + 2 w_{\tau\tau}^\top \Sigma w, \quad (10)$$

and while  $w_\tau = O(\gamma)$  (each appearance of  $\hat{B}$  in (4) carries a factor  $\gamma$ ), the second derivative  $w_{\tau\tau}$  is only  $O(\gamma)$  as well, through the quadratic term visible in (9). The first term of (10) is  $O(\gamma^2)$ ; the second is generically  $O(\gamma)$ . The shortcut “ $w_\tau = O(\gamma)$ , therefore  $F - V_0 = O(\gamma^2\tau^2)$ ” bounds only the first term and is false; Example 1 makes the failure exact.

**Theorem 2** (Noise cost near the HRP end). *Under Assumptions 1 and 2 there is a continuous bounded function  $H$  on  $[0, \bar{\gamma}] \times [0, \bar{\tau}]$  with*

$$F(\gamma; \tau) - V_0(\gamma) = \gamma \tau^2 H(\gamma, \tau), \quad \text{hence} \quad |F(\gamma; \tau) - V_0(\gamma)| \leq C \gamma \tau^2 \quad (11)$$

with  $C = \sup |H|$ . The exponent of  $\gamma$  cannot be improved: in Example 1,  $H(0, 0) = \frac{8}{189} \neq 0$ .

*Proof.* Let  $\Phi(\gamma, \tau) = F(\gamma; \tau) - V_0(\gamma)$ , which is  $C^\infty$  on the compact rectangle  $R = [0, \bar{\gamma}] \times [-\bar{\tau}, \bar{\tau}]$  by Assumption 1. Three exact identities hold on  $R$ :  $\Phi(0, \tau) = 0$  for all  $\tau$  (Lemma 1);  $\Phi(\gamma, 0) = 0$  for all  $\gamma$  (no noise); and  $\Phi(\gamma, -\tau) = \Phi(\gamma, \tau)$  (Assumption 2). Fix  $\gamma$  and Taylor-expand in  $\tau$  with integral remainder: since  $\Phi(\gamma, 0) = 0$  and, by evenness,  $\partial_\tau \Phi(\gamma, 0) = 0$ ,

$$\Phi(\gamma, \tau) = \tau^2 \int_0^1 (1-s) \partial_{\tau\tau}^2 \Phi(\gamma, s\tau) ds =: \tau^2 \Psi(\gamma, \tau).$$

$\Psi$  is  $C^1$  on  $R$  (differentiation under the integral, with the third derivatives of  $\Phi$  bounded on  $R$ ). Differentiating the identity  $\Phi(0, \tau) = 0$  twice in  $\tau$  gives  $\partial_{\tau\tau}^2 \Phi(0, \tau) = 0$  for all  $\tau$ , hence  $\Psi(0, \tau) = 0$ . Taylor in  $\gamma$  with integral remainder then gives

$$\Psi(\gamma, \tau) = \gamma \int_0^1 \partial_\gamma \Psi(u\gamma, \tau) du =: \gamma H(\gamma, \tau),$$

with  $H$  continuous and bounded on  $R$  by the same uniform derivative bounds. Combining the two displays yields (11). The sharpness claim is Example 1, where  $\partial_\gamma(F - V_0)(0; \tau) = \frac{8}{189}\tau^2$  exactly.  $\square$

With  $C^4$  regularity in  $\tau$  (available under Assumption 1), the same identities give the uniform expansion

$$F(\gamma; \tau) = V_0(\gamma) + \tau^2 G(\gamma) + O(\tau^4), \quad G(\gamma) = \frac{1}{2} \partial_{\tau\tau}^2 F(\gamma; 0) = \gamma h(\gamma), \quad (12)$$

with  $h$  continuous, so  $G(0) = 0$ , but in general  $G'(0) = h(0) \neq 0$ :  $G$  vanishes only linearly at the HRP end. The exact boundary derivative is

$$\partial_\gamma F(0; \tau) = V_0'(0) + \tau^2 H(0, \tau). \quad (13)$$

*Remark 4* (When is the stronger  $\gamma^2\tau^2$  bound valid?). The bound  $|F - V_0| \leq C\gamma^2\tau^2$  is recoverable under one additional, substantive hypothesis: that the first coupling variation is unbiased under the noise, i.e.  $\partial_\gamma F(0; \tau) = V_0'(0)$  for all small  $\tau$ . Then both the value and the first  $\gamma$ -derivative of  $F - V_0$  vanish at  $\gamma = 0$  and a third application of the Taylor argument yields  $F - V_0 = \gamma^2\tau^2 \tilde{H}$ . The recursion does not satisfy this generically, because

$$\mathbb{E}[(B + \tau N_B) D^{-1} (B + \tau N_B)^\top] = BD^{-1}B^\top + \tau^2 \mathbb{E}[N_B D^{-1} N_B^\top],$$

and that quadratic bias enters (4) multiplied by a single power of  $\gamma$ . Both exact examples in this paper happen to possess a further downstream cancellation (Remark 7), which is why numerical evidence from them alone can suggest, wrongly, that the quadratic bound is general.

**Example 1** (HRP locally optimal at a finite noise level). *Let  $n = 4$  with*

$$A = D = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{pmatrix}, \quad B = \frac{1}{10} \mathbf{1}\mathbf{1}^\top, \quad E = e_1 e_1^\top, \quad \hat{B} = B \pm \tau E \quad \text{w.p. } \frac{1}{2} \text{ each},$$

and  $\Sigma = \begin{pmatrix} A & B \\ B & A \end{pmatrix}$ , which is SPD (both  $A + B$  and  $A - B$  are positive definite). Then, exactly and for every  $\tau$ ,

$$\partial_\gamma F(0; \tau) = -\frac{1}{700} + \frac{8}{189} \tau^2. \quad (14)$$

In particular  $V'_0(0) = -\frac{1}{700} < 0$ , so the premise of Theorem 1 holds, yet  $\partial_\gamma F(0; \tau) > 0$  as soon as  $\tau^2 > \frac{27}{800}$ , i.e.  $\tau > \sqrt{27/800} \approx 0.1837$ : the HRP endpoint is then a one-sided local minimum. Moreover  $F(\gamma; \tau) - V_0(\gamma) = \frac{8}{189} \gamma \tau^2 + O(\gamma^2)$ , so no bound of the form  $C\gamma^2\tau^2$  can hold on any neighborhood of  $\gamma = 0$ .

*Proof.* The computation is elementary and worth walking through. *Step 1 (symmetry).* Since  $A = D$  and each realized  $\hat{B}$  is symmetric, the two augmented blocks in (4) coincide, so the top-level split is  $\frac{1}{2} : \frac{1}{2}$  for every  $\gamma, \tau$ , and branch, and  $w = \frac{1}{2}(u; u)$  where  $u$  are the inverse-diagonal weights (5) of the common augmented block  $A_\gamma(\hat{B})$ . *Step 2 (objective).* With the true  $\Sigma$ ,

$$V = w^\top \Sigma w = \frac{1}{4}(u^\top A u + 2u^\top B u + u^\top A u) = \frac{1}{2} u^\top (A + B) u.$$

*Step 3 (first-order augmentation).* By (8) with  $D = A$  and  $\hat{B}$  symmetric,  $A_\gamma(\hat{B}) = A + \gamma H(\hat{B}) + O(\gamma^2)$  with  $H(\hat{B}) = \hat{B} - \hat{B} A^{-1} \hat{B}$ ; the symmetrization does not alter this since  $H(\hat{B})$  is symmetric. *Step 4 (weight derivative).* The weights (5) depend only on the diagonal. With  $\text{diag}(A) = (1, 2)$ ,  $u(0) = (\frac{2}{3}, \frac{1}{3})$ , and differentiating  $u(\gamma) = (A_{22} + \gamma H_{22}, A_{11} + \gamma H_{11}) / (3 + \gamma(H_{11} + H_{22}))$  at  $\gamma = 0$  gives

$$u'(0) = \frac{H_{22} - 2H_{11}}{9} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

*Step 5 (variance derivative).* With  $M = A + B$  one computes  $M(1, -1)^\top = (\frac{3}{2}, -\frac{5}{2})^\top$  and  $u(0)^\top M(1, -1)^\top = \frac{2}{3} \cdot \frac{3}{2} - \frac{1}{3} \cdot \frac{5}{2} = \frac{1}{6}$ , so

$$\partial_\gamma V|_{\gamma=0} = u(0)^\top M u'(0) = \frac{H_{22} - 2H_{11}}{54}.$$

*Step 6 (average over branches).*  $H$  is quadratic in  $\hat{B}$ , so averaging is exact:

$$\frac{1}{2} \sum_{\pm} H(B \pm \tau E) = B - B A^{-1} B - \tau^2 E A^{-1} E.$$

With  $A^{-1} = \frac{1}{7} \begin{pmatrix} 8 & 2 \\ 2 & 4 \end{pmatrix}$  one gets  $B A^{-1} B = \frac{4}{175} \mathbf{1}\mathbf{1}^\top$  and  $E A^{-1} E = \frac{8}{7} e_1 e_1^\top$ , so the averaged  $H$  has  $\bar{H}_{11} = \frac{1}{10} - \frac{4}{175} - \frac{8}{7} \tau^2$  and  $\bar{H}_{22} = \frac{1}{10} - \frac{4}{175}$ . Substituting into Step 5,

$$\partial_\gamma F(0; \tau) = \frac{\bar{H}_{22} - 2\bar{H}_{11}}{54} = \frac{-\frac{27}{350} + \frac{16}{7} \tau^2}{54} = -\frac{1}{700} + \frac{8}{189} \tau^2.$$

Since  $F$  is smooth in  $\gamma$  near 0 (denominators reduce to those of  $A$  at  $\gamma = 0$ ), the exchange of expectation and derivative is justified by the finite support of the noise. The certificate script confirms (14) in exact rational arithmetic, by dual-number (first-order jet) evaluation of the full recursion, at  $\tau \in \{0, \frac{1}{10}, \frac{3}{20}, \frac{1}{5}, \frac{3}{10}\}$ : for instance  $\partial_\gamma F(0; \frac{3}{20}) = -\frac{1}{2100}$  and  $\partial_\gamma F(0; \frac{1}{5}) = +\frac{1}{3780}$ .  $\square$

## 4 The minimum-variance end

Suppose the bridge under study reaches minimum variance exactly at  $\gamma = 1$ : writing  $w_* = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{1})$ , suppose  $w(1; \Sigma) = w_*$  while  $\mathbf{1}^\top w(\gamma; \Sigma) = 1$  for all  $\gamma$ . This holds for the canonical member of the family (Proposition 1) and, in the solved family of Section 5, for the implemented variant as well; it is not automatic for the implemented variant in general.

**Lemma 2** (Flatness at an exact minimum-variance endpoint). *Under the above,  $V_0$  is differentiable at  $\gamma = 1$  with*

$$V'_0(1) = 0, \quad V''_0(1) = 2w_\gamma(1)^\top \Sigma w_\gamma(1) \geq 0,$$

where  $w_\gamma(1) = \partial_\gamma w(\gamma; \Sigma)|_{\gamma=1}$ ; the second derivative is strictly positive whenever the bridge arrives with a nonzero tangent,  $w_\gamma(1) \neq 0$ .

*Proof.* Since  $\Sigma w_* = \lambda \mathbf{1}$  with  $\lambda = 1/(\mathbf{1}^\top \Sigma^{-1}\mathbf{1})$ , differentiating  $V_0(\gamma) = w(\gamma)^\top \Sigma w(\gamma)$  at  $\gamma = 1$  gives  $V'_0(1) = 2w_\gamma(1)^\top \Sigma w_* = 2\lambda w_\gamma(1)^\top \mathbf{1} = 0$ , because  $\mathbf{1}^\top w(\gamma) \equiv 1$  implies  $\mathbf{1}^\top w_\gamma \equiv 0$ . Differentiating once more,  $V''_0(1) = 2w_\gamma(1)^\top \Sigma w_\gamma(1) + 2w_{\gamma\gamma}(1)^\top \Sigma w_*$ , and the second term vanishes for the same reason ( $\mathbf{1}^\top w_{\gamma\gamma} \equiv 0$ ).  $\square$

**Theorem 3** (Small noise moves the optimum inside). *Assume the setting of Lemma 2 with  $V''_0(1) > 0$ , the expansion (12) valid in  $C^2$  uniformly near  $\gamma = 1$ , and  $G'(1) > 0$ . Then there are  $\tau_1 > 0$  and a  $C^1$  curve  $\gamma^*(\tau)$ , defined for  $0 \leq \tau < \tau_1$ , of local minimizers of  $F(\cdot; \tau)$  with*

$$\gamma^*(\tau) = 1 - \frac{G'(1)}{V''_0(1)} \tau^2 + O(\tau^4) < 1 \quad (\tau > 0). \quad (15)$$

*If moreover  $\gamma = 1$  is the unique global minimizer of  $V_0$  on  $[0, \bar{\gamma}]$ , then for all sufficiently small  $\tau$  the curve gives the unique global minimizer of  $F(\cdot; \tau)$ . The sign condition is essential: if  $G'(1) < 0$ , full coupling remains locally optimal for small noise.*

*Proof.* Let  $g(\gamma, s) = \partial_\gamma F(\gamma; \sqrt{s})$  for  $s = \tau^2 \geq 0$ ; by (12) in  $C^2$ ,  $g(\gamma, s) = V'_0(\gamma) + s G'(\gamma) + O(s^2)$  with the error uniform in  $\gamma$  near 1, and  $g$  is  $C^1$  in  $(\gamma, s)$  there. By Lemma 2,  $g(1, 0) = V'_0(1) = 0$  and  $\partial_\gamma g(1, 0) = V''_0(1) > 0$ . The implicit function theorem gives a  $C^1$  curve  $\gamma^*(s)$  with  $g(\gamma^*(s), s) = 0$ ,  $\gamma^*(0) = 1$ , and

$$\left. \frac{d\gamma^*}{ds} \right|_{s=0} = - \frac{\partial_s g(1, 0)}{\partial_\gamma g(1, 0)} = - \frac{G'(1)}{V''_0(1)} < 0,$$

which is (15). Since  $\partial_\gamma g > 0$  near  $(1, 0)$ , each  $\gamma^*(s)$  is a strict local minimizer. For global optimality:  $F(\cdot; \tau) \rightarrow V_0$  uniformly on  $[0, \bar{\gamma}]$  as  $\tau \rightarrow 0$  (by (11) or directly by uniform continuity), so if  $\gamma = 1$  is the unique global minimizer of  $V_0$ , minimizers of  $F$  must enter any prescribed neighborhood of 1 for small  $\tau$  (outside such a neighborhood  $V_0$  exceeds  $V_0(1)$  by a positive margin, by compactness), where strict convexity of  $F$  in  $\gamma$  (from  $\partial_\gamma g > 0$ ) makes  $\gamma^*(s)$  the only candidate. If  $G'(1) < 0$  the same computation gives  $d\gamma^*/ds > 0$ : the unconstrained stationary point exits the interval at  $\gamma = 1$  and full coupling is locally optimal.  $\square$

## 5 An exactly solved family

Fix the four-asset covariance

$$\Sigma = \begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{5} & \frac{3}{5} \\ \frac{1}{2} & 1 & \frac{3}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{3}{5} & 4 & 2 \\ \frac{3}{5} & \frac{3}{5} & 2 & 4 \end{pmatrix}, \quad (16)$$



i.e. blocks  $\{1, 2\}$  and  $\{3, 4\}$  with unit and double volatilities, within-block correlation  $\frac{1}{2}$ , and cross-block correlation  $\rho_c = \frac{3}{10}$ . Let the noise perturb the cross-block correlation *in common*: all four cross entries move together,  $\rho_c \rightarrow \rho_c \pm \tau$  with probability  $\frac{1}{2}$  each. At the level of the two blocks' equal-weight portfolios, the population variances and cross covariance are

$$a = \frac{3}{4}, \quad d = 3, \quad c = \frac{3}{5}, \quad S := a + d = \frac{15}{4}, \quad K := S - 2c = \frac{51}{20},$$

and the estimated group-level cross covariance is  $Z = c \pm e$  with  $e = 2\tau$  (the blocks have volatilities 1 and 2).

**Lemma 3** (Exact reduction to a scalar bridge). *Assume  $a < d$  and  $4c^2 < ad$ , as in (16), and fix  $\gamma \in [0, 1]$  and a realized  $z = c \pm e$  with  $0 \leq z \leq 2c$ . The augmentation (4) of the  $D$  block requires the factor  $I - \gamma B^\top A^{-1}$  to be invertible, and that holds precisely when  $\gamma z \neq a$ ; the  $A$ -side factor  $I - \gamma B D^{-1}$  is invertible for every such  $(\gamma, z)$ , since  $\gamma z \leq 2c < d$ . Wherever the recursion is defined, its within-block weights are exactly  $(\frac{1}{2}, \frac{1}{2})$  in each block, the augmented fitnesses are*

$$\nu_{A_\gamma}(z) = \frac{ad - \gamma z^2}{d - \gamma z}, \quad \nu_{D_\gamma}(z) = \frac{ad - \gamma z^2}{a - \gamma z}, \quad (17)$$

and the left-block weight is

$$x_\gamma(z) = \frac{d - \gamma z}{a + d - 2\gamma z}, \quad (18)$$

with realized out-of-sample variance  $Q(x_\gamma(z))$ , where  $Q(x) = ax^2 + d(1-x)^2 + 2cx(1-x)$ . The formula (18) is a rational function of  $(\gamma, z)$  whose denominator obeys  $S - 2\gamma z \geq S - 4c > 0$ . It therefore extends the recursion's group weight continuously through the point  $\gamma z = a$ , where the extension gives the  $D$  block the weight  $1 - x_\gamma(z) = (a - \gamma z)/(S - 2\gamma z) = 0$ . This is the analog of the  $b = 0$  convention in Proposition 1: a block whose augmented fitness diverges receives nothing. We call the allocation defined by (18) on all of  $[0, 1]$  the extended allocation and write  $F_{\text{ext}}$  for its expected out-of-sample variance. For the numerical family (16) the ordering  $c = \frac{3}{5} < \frac{3}{4} = a$  holds, so the lower branch has  $z = c - e < a$  throughout and never meets  $\gamma z = a$ ; the sole possible singularity is on the upper branch, at  $\gamma = a/(c+e)$ , and it lies in  $[0, 1]$  precisely when  $e \geq a - c = \frac{3}{20}$ . Moreover, for (16) every other denominator the recursion divides by (the augmented diagonal entries and the fitness sum) is nonzero on the admissible range, so  $\gamma z = a$  is the only point at which the recursion is undefined. Without  $c < a$  both branches can be singular;  $a = 1$ ,  $d = 6$ ,  $c = \frac{11}{10}$  satisfies  $a < d$ ,  $4c^2 < ad$ , and  $S > 6c$ , yet at  $e = \frac{1}{20}$  both  $\gamma = \frac{20}{21}$  (lower branch) and  $\gamma = \frac{20}{23}$  (upper branch) are singular.

*Proof.* Write the asset-level blocks as  $A_2 = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$ ,  $D_2 = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$ , and  $\hat{B} = z \mathbf{1}\mathbf{1}^\top$  (each cross entry equals  $1 \cdot 2 \cdot (\rho_c \pm \tau) = z$ ). All three matrices are exchangeable (invariant under swapping the two assets of a block), and the augmentation (4) preserves exchangeability, so each augmented block has equal diagonal entries and (5) gives  $(\frac{1}{2}, \frac{1}{2})$  within blocks. Invertibility first. Since  $A_2^{-1} \mathbf{1} = \mathbf{1}/(2a)$ , the  $D$ -side factor is  $I - \gamma B^\top A_2^{-1} = I - \gamma \frac{z}{2a} \mathbf{1}\mathbf{1}^\top$ , whose eigenvalues are 1 (on the difference vector) and  $1 - \gamma z/a$  (on  $\mathbf{1}$ ); it is singular precisely when  $\gamma z = a$ . The  $A$ -side factor  $I - \gamma \frac{z}{2d} \mathbf{1}\mathbf{1}^\top$  has eigenvalue  $1 - \gamma z/d > 0$ , because  $\gamma z \leq 2c < d$ ; here  $2c < d$  follows from  $4c^2 < ad \leq d^2$ . The common numerator in (17) also never vanishes:  $ad - \gamma z^2 \geq ad - 4c^2 > 0$ . The matrix factors are not the only denominators. The recursion also divides by the augmented diagonal entries and by the fitness sum, and exchangeability settles both. The difference vector  $v = (1, -1)^\top$  satisfies  $\mathbf{1}\mathbf{1}^\top v = 0$ , so  $\hat{B}v = \hat{B}^\top v = 0$ : the Schur correction in (4) annihilates  $v$  and the collapse factor fixes  $v$ , hence each augmented block keeps its population difference eigenvalue,  $\frac{1}{2}$  for the  $A$  block ( $1 - \frac{1}{2}$ )

and 2 for the  $D$  block ( $4 - 2$ ). An exchangeable  $2 \times 2$  matrix has both diagonal entries equal to the average of its two eigenvalues, and its  $\mathbf{1}$  eigenvalue is twice its fitness, so

$$(A_\gamma)_{ii} = \nu_{A_\gamma}(z) + \frac{1}{4}, \quad (D_\gamma)_{ii} = \nu_{D_\gamma}(z) + 1 = \frac{a(d+1) - \gamma z(z+1)}{a - \gamma z}.$$

The first is positive outright, since  $\nu_{A_\gamma}(z) > 0$  by (17). The second has numerator at least  $a(d+1) - 2c(2c+1) = 3 - \frac{66}{25} = \frac{9}{25} > 0$  for (16), so it can vanish nowhere and blows up only through the factor  $a - \gamma z$  already identified. Finally

$$\nu_{A_\gamma}(z) + \nu_{D_\gamma}(z) = \frac{(ad - \gamma z^2)(S - 2\gamma z)}{(d - \gamma z)(a - \gamma z)},$$

whose numerator is positive on the admissible range; so for (16) the point  $\gamma z = a$  is the recursion's only hole. For (17): since  $D_2$  is exchangeable with row sums  $2d$ ,  $D_2^{-1}\mathbf{1} = \mathbf{1}/(2d)$ , so  $\widehat{B}D_2^{-1} = \frac{z}{2d}\mathbf{1}\mathbf{1}^\top$  and  $\widehat{B}D_2^{-1}\widehat{B}^\top = \frac{z^2}{d}\mathbf{1}\mathbf{1}^\top$  (using  $\mathbf{1}\mathbf{1}^\top\mathbf{1}\mathbf{1}^\top = 2\mathbf{1}\mathbf{1}^\top$ ). The matrix  $r = I - \gamma\frac{z}{2d}\mathbf{1}\mathbf{1}^\top$  satisfies  $\mathbf{1}^\top r^{-1} = (1 - \gamma z/d)^{-1}\mathbf{1}^\top$  (Sherman–Morrison), and sym does not change  $\mathbf{1}^\top(\cdot)\mathbf{1}$ . With within weights  $(\frac{1}{2}, \frac{1}{2})$ , the fitness is  $\nu_{A_\gamma} = \frac{1}{4}\mathbf{1}^\top A_\gamma \mathbf{1} = \frac{1}{4} \frac{\mathbf{1}^\top A_2 \mathbf{1} - \gamma \frac{z^2}{d} \cdot 4}{1 - \gamma z/d} = \frac{a - \gamma z^2/d}{1 - \gamma z/d} = \frac{ad - \gamma z^2}{d - \gamma z}$ , using  $\mathbf{1}^\top A_2 \mathbf{1} = 4a$ ; the  $D$ -side is symmetric. For (18): the inter-group ratio is  $1/\nu_{A_\gamma} : 1/\nu_{D_\gamma}$ , the common factor  $ad - \gamma z^2$  cancels, and  $x_\gamma(z) = (d - \gamma z)/((d - \gamma z) + (a - \gamma z))$ . The final sentence is the definition of  $Q$  at the group level. Each identity is also verified exactly by the certificate script at a grid of rational  $(\gamma, z)$  away from the singular point; at the singular point the script verifies that the recursion is undefined (a division by an exact zero) and that the recursion's weight approaches (18) on both sides.  $\square$

**Proposition 2** (Exact objective). *Let  $x_* = x_1(c) = (d - c)/K$  be the population minimum-variance group weight. Then  $Q(x) = Q(x_*) + K(x - x_*)^2$  (exact square completion),*

$$x_\gamma(z) - x_* = \frac{(d - a)(\gamma z - c)}{(S - 2\gamma z)K}, \quad (19)$$

and therefore, with  $r(u) = ((u - c)/(S - 2u))^2$ ,

$$F_{\text{ext}}(\gamma; e) = Q(x_*) + \frac{(d - a)^2}{K} \cdot \frac{1}{2} \sum_{\sigma=\pm 1} r(\gamma(c + \sigma e)). \quad (20)$$

*This is the exact expected out-of-sample variance of the extended allocation; no Taylor surrogate is involved. It agrees with the objective of the raw recursion at every  $(\gamma, e)$  where the recursion is defined; by Lemma 3, for the numerical family (16) that is everywhere except the single point  $\gamma = a/(c + e)$  when  $e \geq a - c$ . In particular*

$$V_0(\gamma) = \frac{9(44\gamma^2 - 200\gamma + 275)}{5(8\gamma - 25)^2}, \quad G(\gamma) = \frac{1500\gamma^2(16\gamma + 1)}{(8\gamma - 25)^4}, \quad (21)$$

where  $G$  is the coefficient of  $e^2$  in the small-noise expansion of  $F_{\text{ext}}(\gamma; e)$ ; for  $e < a - c$  there is no singular point and  $F_{\text{ext}}$  is the raw objective outright.

*Proof.*  $Q$  is a quadratic in  $x$  with leading coefficient  $a + d - 2c = K$  and vertex at  $x_*$  (set  $Q'(x) = 2ax - 2d(1 - x) + 2c(1 - 2x) = 2Kx - 2(d - c)$ ), giving the square completion. For (19), put both

terms of  $x_\gamma(z) - x_*$  over the common denominator  $(S - 2\gamma z)K$ : the numerator is  $(d - \gamma z)K - (d - c)(S - 2\gamma z) = (\gamma z - c)(2d - S) = (d - a)(\gamma z - c)$ , expanding and using  $K = S - 2c$ . Then  $Q(x_\gamma(z)) - Q(x_*) = K(x_\gamma(z) - x_*)^2 = \frac{(d-a)^2}{K} r(\gamma z)$ , and averaging over  $z = c \pm e$  gives (20). The closed forms (21) follow by substituting the family's numbers into (20); they are verified exactly, at a grid of rational  $\gamma$ , by the certificate script.  $\square$

**Theorem 4** (Exact interior optimum, monotone in the noise). *Assume  $a < d$ ,  $c > 0$ ,  $4c^2 < ad$ , and  $S > 6c$  (all satisfied by (16):  $4c^2 = \frac{36}{25} < \frac{9}{4} = ad$  and  $S = \frac{15}{4} > \frac{18}{5} = 6c$ ). Then for every noise amplitude  $0 < e \leq c$ , with  $F_{\text{ext}}$  the extended objective of Lemma 3:*

- (i)  $F_{\text{ext}}(\cdot; e)$  is strictly convex on  $[0, 1]$ ;
- (ii) it has a unique minimizer  $\gamma^*(e) \in (\frac{1}{2}, 1)$  for  $0 < e < c$ , with the boundary values  $\gamma^*(0) = 1$  and  $\gamma^*(c) = \frac{1}{2}$ ;
- (iii)  $\gamma^*$  is strictly decreasing in  $e$ ;
- (iv) the minimizer solves the exact first-order condition

$$\sum_{\sigma=\pm 1} \frac{(c + \sigma e) [\gamma(c + \sigma e) - c]}{[S - 2\gamma(c + \sigma e)]^3} = 0, \quad (22)$$

and for small  $e$ ,

$$\gamma^*(e) = 1 - \frac{S + 4c}{c^2(S - 2c)} e^2 + O(e^4) = 1 - \frac{1025}{153} e^2 + O(e^4) \quad (23)$$

for the numbers of (16);

- (v) for the numbers of (16) the minimizer lies strictly inside the regular region: whenever the singular point  $\gamma_e = a/(c + e)$  of Lemma 3 lies in  $[0, 1]$ , that is for  $e \geq a - c = \frac{3}{20}$ ,

$$\gamma^*(e) < \frac{a}{c + e}.$$

Consequently the raw recursion is defined on a neighborhood of  $\gamma^*(e)$  and coincides there with  $F_{\text{ext}}$ , so (i)–(iv) describe the optimizer of the implemented recursion and not only of the extension.

Consequently every coupling in  $[\frac{1}{2}, 1]$  is optimal at exactly one noise amplitude  $e \in [0, c]$ . In the noise parametrization of the paper,  $e = 2\tau$ , so the theorem covers  $0 < \tau \leq \frac{3}{10}$ .

*Proof.* Throughout, drop the positive constant  $(d - a)^2/K$  and study  $\bar{F}(\gamma; e) = \frac{1}{2}[r(\gamma(c + e)) + r(\gamma(c - e))]$  with  $r(u) = (u - c)^2/(S - 2u)^2$ , defined for  $u < S/2$ ; note  $\gamma z \leq 2c < S/2$  on the admissible range, since  $z \leq c + e \leq 2c$  and  $S > 6c > 4c$ . By (20),  $\bar{F}$  is  $F_{\text{ext}}$  up to a positive affine map, and it is smooth on all of  $[0, 1] \times [0, c]$ ; the point  $\gamma z = a$ , where the raw recursion is undefined, is unremarkable for  $\bar{F}$ .

*Derivatives of  $r$ .* Direct differentiation gives

$$r'(u) = \frac{2K(u - c)}{(S - 2u)^3}, \quad r''(u) = \frac{2K(S - 6c + 4u)}{(S - 2u)^4}.$$

(For  $r'$ : the quotient rule gives numerator  $2(u - c)(S - 2u) + 4(u - c)^2 = 2(u - c)(S - 2c)$ ; for  $r''$ , differentiate  $r'$  and simplify the numerator to  $2K[(S - 2u) + 6(u - c)]$ .)

(i) *Convexity.*  $\partial_{\gamma\gamma}^2 \bar{F}(\gamma; e) = \frac{1}{2} \sum_{\sigma} z_{\sigma}^2 r''(\gamma z_{\sigma})$  with  $z_{\sigma} = c + \sigma e \geq 0$ . Since  $S > 6c$  and  $u = \gamma z_{\sigma} \geq 0$ , the numerator  $S - 6c + 4u$  is strictly positive, and  $(S - 2u)^4 > 0$ , so  $r'' > 0$  wherever  $z_{\sigma} > 0$ ; at  $e = c$  the lower branch has  $z_- = 0$  and contributes nothing, while the upper branch still gives strict positivity. Hence  $\bar{F}$  is strictly convex in  $\gamma$  on  $[0, 1]$ .

*Endpoint signs.* At  $\gamma = 0$ :  $\partial_{\gamma} \bar{F}(0; e) = \frac{1}{2} \sum_{\sigma} z_{\sigma} r'(0) = r'(0) c = -2Kc^2/S^3 < 0$ , independently of  $e$ . At  $\gamma = 1$ : define  $\psi(z) = z r'(z) = 2Kz(z - c)/(S - 2z)^3$ , so  $\partial_{\gamma} \bar{F}(1; e) = \frac{1}{2}[\psi(c + e) + \psi(c - e)]$ . One computes

$$\psi''(z) = \frac{4K [S(S - 6c) + 4z(2S - 3c) + 4z^2]}{(S - 2z)^5} > 0 \quad (0 \leq z \leq 2c),$$

since every bracketed term is positive under  $S > 6c$  and  $S - 2z \geq S - 4c > 0$ . Thus  $\psi$  is strictly convex on  $[0, 2c]$ , and since  $\psi(c) = 0$ , strict midpoint convexity gives  $\frac{1}{2}[\psi(c + e) + \psi(c - e)] > \psi(c) = 0$  for  $0 < e \leq c$ . So the  $\gamma$ -derivative is strictly negative at  $\gamma = 0$  and strictly positive at  $\gamma = 1$ .

(ii) *Unique interior minimizer.* Strict convexity plus the endpoint signs give a unique root  $\gamma^*(e) \in (0, 1)$  of  $\partial_{\gamma} \bar{F} = 0$ , which is the unique minimizer, and (22) is  $\partial_{\gamma} \bar{F} = 0$  written out. At  $e = c$  the lower branch has  $z_- = 0$ , so (22) reduces to  $r'(2c\gamma) = 0$ , whose unique solution is  $2c\gamma = c$ , i.e.  $\gamma^*(c) = \frac{1}{2}$ ; and  $\gamma^*(e) > \frac{1}{2}$  for  $e < c$  follows from (iii). At  $e = 0$  both branches coincide,  $\partial_{\gamma} \bar{F} \propto r'(\gamma c) c$ , whose root is  $\gamma = 1$ ; this is the noise-free optimum, consistent with  $V_0'(1) = 0$  (Lemma 2 applies: the reduction reaches the group-level minimum variance at  $\gamma = 1$  exactly).

(iii) *Monotonicity.* For fixed  $\gamma > 0$  write  $\psi_{\gamma}(z) = z r'(\gamma z)$ , so that  $\partial_{\gamma} \bar{F}(\gamma; e) = \frac{1}{2}[\psi_{\gamma}(c + e) + \psi_{\gamma}(c - e)]$  and

$$\partial_e \partial_{\gamma} \bar{F}(\gamma; e) = \frac{1}{2}[\psi'_{\gamma}(c + e) - \psi'_{\gamma}(c - e)].$$

The identity  $\psi_{\gamma}(z) = \psi(\gamma z)/\gamma$  gives  $\psi''_{\gamma}(z) = \gamma \psi''(\gamma z) > 0$  on the admissible range, so  $\psi'_{\gamma}$  is strictly increasing,  $\psi'_{\gamma}(c + e) > \psi'_{\gamma}(c - e)$ , and the mixed partial is strictly positive. By the implicit function theorem on (22),  $d\gamma^*/de = -\partial_e \partial_{\gamma} \bar{F} / \partial_{\gamma\gamma}^2 \bar{F} < 0$ .

(iv) *Expansion.* Insert  $\gamma = 1 - \delta$  into (22) and expand to first order in  $\delta$  and  $e^2$ . The  $e$ -expansion of the sum at  $\gamma = 1$  is  $\frac{2e^2(S+4c)}{K^4} + O(e^4)$  (expand each branch of (22) in  $e$ ; odd terms cancel), while the  $\delta$ -derivative at  $e = 0$  is  $-2c^2/K^3 + O(\delta, e^2)$ . Setting the sum of the two contributions to zero gives  $\delta = e^2(S + 4c)/(c^2 K) + O(e^4)$ , which is (23); the coefficient evaluates to  $\frac{1025}{153}$  for (16).

(v) *The singular point lies uphill.* Let  $\gamma_e = a/(c + e) \leq 1$  and write  $z_{\pm} = c \pm e$ , so that  $\gamma_e z_+ = a$  and  $\gamma_e z_- = az_-/z_+$ . By strict convexity it suffices to show  $\partial_{\gamma} \bar{F}(\gamma_e; e) > 0$ . Substituting into  $\partial_{\gamma} \bar{F} = \frac{1}{2} \sum_{\sigma} z_{\sigma} r'(\gamma z_{\sigma})$  and clearing the inner fractions,

$$\partial_{\gamma} \bar{F}(\gamma_e; e) = K \left[ \frac{z_+(a - c)}{(S - 2a)^3} + \frac{z_-(az_- - cz_+) z_+^2}{(Sz_+ - 2az_-)^3} \right].$$

Both denominators are positive:  $S - 2a = \frac{9}{4}$  for (16), and  $Sz_+ - 2az_- \geq z_+(S - 2a) > 0$  since  $z_- \leq z_+$ . Multiplying by the positive quantity  $(S - 2a)^3(Sz_+ - 2az_-)^3/K$  preserves the sign and produces the polynomial

$$P(e) = z_+(a - c)(Sz_+ - 2az_-)^3 + z_-(az_- - cz_+) z_+^2 (S - 2a)^3,$$

which for the numbers of (16) expands to

$$P(e) = \frac{177147}{400000} + \frac{6561}{800} e^2 + \frac{1215}{32} e^3 + \frac{23733}{640} e^4.$$

Every coefficient is nonnegative and the constant term is positive, so  $P(e) > 0$  for every  $e \geq 0$ , and in particular on  $[\frac{3}{20}, \frac{3}{5}]$ . Hence  $\partial_{\gamma} \bar{F}(\gamma_e; e) > 0$ , and by strict convexity  $\gamma^*(e) < \gamma_e$ . The expansion

of  $P$  and the sign agreement with the first-order condition are verified in exact rational arithmetic by the certificate script.

The final sentence of the theorem follows from (ii)–(iii):  $\gamma^* : [0, c] \rightarrow [\frac{1}{2}, 1]$  is a continuous strictly decreasing bijection.  $\square$

**Corollary 1** (The surrogate frontier spans the whole bridge in this family). *For the family (16), the closed forms (21) satisfy  $V'_0 < 0$ ,  $V''_0 > 0$ ,  $G' > 0$ ,  $G'' > 0$  on  $(0, 1)$ , and*

$$t(\gamma) = -\frac{V'_0(\gamma)}{G'(\gamma)} = \frac{9(1-\gamma)(25-8\gamma)^2}{25\gamma(64\gamma^2+608\gamma+25)}$$

*decreases strictly from  $+\infty$  (at  $\gamma \downarrow 0$ ) to 0 (at  $\gamma \uparrow 1$ ). In the language of Theorem 5 below,  $t_0 = \infty$  and  $t_1 = 0$ : for this family, and only because of these endpoint limits, the surrogate optimizer sweeps the entire open bridge and neither endpoint is surrogate-optimal at any positive finite noise level. The surrogate optimizer is not the exact optimizer at finite noise. At  $e = c$  the exact minimizer is  $\frac{1}{2}$  (Theorem 4), while  $t(\frac{1}{2}) = \frac{1323}{2875} \neq e^2 = \frac{9}{25}$  and the surrogate optimizer sits near 0.537. The two agree to order  $e^2$  as  $e \rightarrow 0$  (Proposition 3); what they share exactly is the sweep of the whole open bridge.*

*Remark 5.* Theorem 4 reproduces, exactly, the numerical optimizer sweep previously obtained by brute force on this family: with  $e = 2\tau$ , the exact minimizers at  $\tau = 0.10, 0.15, 0.20, 0.25, 0.30$  are  $\gamma^* \approx 0.8308, 0.7237, 0.6339, 0.5603, 0.5000$ , the last being exact. The value  $\tau = 0.35$  falls outside the range  $e \leq c$  covered by the theorem (one noise branch then has negative estimated cross covariance); it could be handled by direct polynomial certification but we do not pursue it.

## 6 The surrogate in general

Away from solvable families one can still organize the problem through the second-order surrogate  $F_2 = V_0 + \tau^2 G$  of (12).

**Theorem 5** (Clipped monotone frontier, for the surrogate). *Suppose  $V_0$  and  $G$  are continuous on  $[0, \bar{\gamma}]$  and  $C^2$  on  $(0, \bar{\gamma})$ , with:  $V'_0 < 0$ ;  $G' > 0$ ;  $V''_0 \geq 0$  and  $G'' \geq 0$ ; and at every  $\gamma$  at least one of  $V''_0(\gamma) > 0$ ,  $G''(\gamma) > 0$  holds. Define  $t = -V'_0/G'$  on  $(0, \bar{\gamma})$  and its endpoint limits  $t_0 = \lim_{\gamma \downarrow 0} t(\gamma) \in (0, \infty]$  and  $t_1 = \lim_{\gamma \uparrow \bar{\gamma}} t(\gamma) \in [0, \infty)$ . Then  $t$  is continuous and strictly decreasing, and for every  $s = \tau^2 > 0$  the surrogate  $F_2(\cdot; \tau)$  has the unique minimizer*

$$\gamma_2^*(s) = \begin{cases} \bar{\gamma}, & s \leq t_1, \\ t^{-1}(s), & t_1 < s < t_0, \\ 0, & s \geq t_0, \end{cases} \quad (24)$$

*which is nonincreasing in  $s$ . In particular the endpoints are optimal for the surrogate on nontrivial intervals of noise levels precisely when  $t_1 > 0$ , respectively  $t_0 < \infty$ . If moreover  $V_0, G \in C^1([0, \bar{\gamma}])$  with  $V'_0(0) < 0$ , as holds under Assumption 1 via (12), then  $t_0 = -V'_0(0)/G'(0)$ , finite exactly when  $G'(0) > 0$ . Endpoint regularity is needed for that formula:  $V_0(\gamma) = V_0(0) - \sqrt{\gamma}$  near 0, with smooth  $G$ , satisfies every open-interval hypothesis and forces  $t_0 = \infty$  regardless of  $G'(0)$ . In Example 1,  $t_0 = \frac{27}{800}$ ; in the family of Section 5,  $t_0 = \infty$  and  $t_1 = 0$  (Corollary 1).*

*Proof.* Strict monotonicity of  $t$ : on  $(0, \bar{\gamma})$ ,

$$t' = \frac{-V''_0 G' + V'_0 G''}{(G')^2},$$

where the first numerator term is  $\leq 0$  (since  $V_0'' \geq 0$ ,  $G' > 0$ ) and the second is  $\leq 0$  (since  $V_0' < 0$ ,  $G'' \geq 0$ ); at each  $\gamma$  at least one is strictly negative by hypothesis, so  $t' < 0$  pointwise, and  $t$  is a continuous strictly decreasing bijection from  $(0, \bar{\gamma})$  onto  $(t_1, t_0)$ . Sign analysis of  $F_2' = V_0' + sG'$ : since  $G' > 0$ ,  $F_2'(\gamma) < 0 \iff s < t(\gamma)$ . If  $t_1 < s < t_0$  then  $F_2' < 0$  on  $(0, t^{-1}(s))$  and  $F_2' > 0$  on  $(t^{-1}(s), \bar{\gamma})$ , so  $t^{-1}(s)$  is the unique minimizer. If  $s \leq t_1$  then  $s < t(\gamma)$  for every interior  $\gamma$ , so  $F_2' < 0$  throughout and the minimum sits at  $\bar{\gamma}$ ; if  $s \geq t_0$  then  $F_2' > 0$  throughout and the minimum sits at 0. Monotonicity of (24) in  $s$  follows from the monotonicity of  $t^{-1}$  and the ordering of the three regimes.  $\square$

**Proposition 3** (Local transfer to the exact objective). *Fix  $\tau$  and let  $I \subseteq [0, \bar{\gamma}]$  be an interval containing  $\gamma_2^*(\tau^2)$  on which  $F_2'' \geq c > 0$  and  $|F - F_2| \leq K\tau^4$ . If  $\gamma_F$  minimizes  $F$  over  $I$ , then*

$$|\gamma_F - \gamma_2^*| \leq 2\sqrt{K/c} \tau^2.$$

*No claim is made beyond  $I$ : uniqueness, global optimality, and the monotonicity of Theorem 5 are statements about the surrogate.*

*Proof.* Strong convexity of  $F_2$  on the interval  $I$  gives, for the minimizer  $\gamma_2^*$  of  $F_2$ ,  $\frac{c}{2}(\gamma_F - \gamma_2^*)^2 \leq F_2(\gamma_F) - F_2(\gamma_2^*)$ . Replace  $F_2$  by  $F$  twice, paying  $K\tau^4$  each time:  $F_2(\gamma_F) - F_2(\gamma_2^*) \leq F(\gamma_F) - F(\gamma_2^*) + 2K\tau^4$ . Finally  $F(\gamma_F) - F(\gamma_2^*) \leq 0$  because  $\gamma_F$  minimizes  $F$  over  $I$  and  $\gamma_2^* \in I$ . Combine and solve for  $|\gamma_F - \gamma_2^*|$ .  $\square$

*Remark 6* (The same dial in meteorology). The trade-off studied here is not particular to finance. The damped Schur complement (1) coincides, term for term, with the reliability shrinkage that keeps Gaussian (Vecchia) pseudo-likelihoods of high-dimensional weather fields estimable when stations outnumber observations [8]. In that language  $V_0$  is the loss of a spatial model that trusts its estimated conditional structure, and the clipped frontier of Theorem 5 says how much conditional structure to trust at a given sampling depth. The scale  $\tau$  is estimation error, not depth itself;  $\tau^2$  is proportional to inverse effective sampling depth, as in the  $\tau \asymp T^{-1/2}$  correspondence of Remark 2. The spatial literature also fits the damping intensity from data, which supplies a practical estimate of the noise level  $s = \tau^2$  at which to read (24); the correspondence and a comparison of recipes are developed in [8].

## 7 A one-entry example

The solved family relies on the exchangeability of common cross-block noise. The next example removes that symmetry: the noise sits in a single covariance entry, and the endpoint comparisons are certified exactly.

**Example 2** (An interior minimizer, exactly, without common-noise symmetry). *Take the covariance (16) and let the estimate err in the (1, 3) entry only, by  $\pm \frac{3}{20} \cdot 2$  with probability  $\frac{1}{2}$  each. Then, in exact rational arithmetic,*

$$F(\frac{9}{10}) < F(0) \quad (\text{margin } 4.879 \times 10^{-2}), \quad F(\frac{9}{10}) < F(1) \quad (\text{margin } 5.702 \times 10^{-4}),$$

*so every minimizer of  $F$  on  $[0, 1]$  is strictly interior. Both inequalities are strict, and the certified denominator bounds hold uniformly on  $[0, 1]$  and vary continuously with the covariance and noise parameters, so the same conclusion holds on an open neighborhood of these parameters: the phenomenon is robust, not a knife-edge rational coincidence. This is also the minimal non-trivial configuration: because nodes with a singleton child perform no augmentation,  $n \leq 3$  admits no*

$\gamma$ -dependence at all, and  $n = 4$  in two blocks of two is the smallest universe on which the bridge exists.

*Proof.*  $F(0)$ ,  $F(\frac{9}{10})$ ,  $F(1)$  are exact rational numbers, each a finite composition of field operations (4) on rational inputs, and the two inequalities are verified in exact fraction arithmetic by the certificate script. Three values settle the whole interval as follows. Every denominator arising in the recursion is a polynomial in  $\gamma$  with rational coefficients; the script certifies each strictly positive on  $[0, 1]$ , on both noise branches, by exhibiting positive Bernstein coefficients (with exact subdivision where needed). Hence  $F$  is continuous on  $[0, 1]$  and attains its minimum, and  $\min F \leq F(\frac{9}{10}) < \min\{F(0), F(1)\}$ , so no minimizer is an endpoint. The same positivity gives the positive definiteness of the augmented blocks for every  $\gamma$ . Robustness: the certificate proves more than positivity at three points. Each denominator polynomial is strictly positive on all of  $[0, 1]$ , so its minimum over  $[0, 1]$  is positive. The coefficients of these polynomials are themselves polynomial in the covariance entries and the noise amplitude, and the minimum of a polynomial over a compact interval is a continuous function of its coefficients, so every denominator stays uniformly positive on an open neighborhood of the stated parameters. On that neighborhood  $F$  remains continuous on all of  $[0, 1]$  and attains its minimum, the three values  $F(0)$ ,  $F(\frac{9}{10})$ ,  $F(1)$  vary continuously, and both strict inequalities persist: every minimizer stays interior.  $\square$

*Remark 7* (A structural cancellation, special to these families). In both noise models on the covariance (16) — the common cross-correlation noise of Section 5 and the one-entry noise of Example 2 — the boundary derivative does not move with the noise: exact jet evaluation gives

$$\partial_\gamma F(0; \tau) = V'_0(0) = -\frac{216}{3125} \quad \text{exactly, at } \tau \in \{\frac{1}{10}, \frac{3}{20}, \frac{1}{5}\},$$

in both models. In the notation of (12),  $G'(0) = 0$  for these families; for the solved family this is also visible in (21), where  $G(\gamma) \propto \gamma^2(16\gamma + 1)$  vanishes quadratically. That is a cancellation particular to these families, not a general fact: Example 1 has  $G'(0) = \frac{8}{189} > 0$ . It explains why numerical experiments on these families alone can suggest, wrongly, that the noise cost is  $O(\gamma^2\tau^2)$  in general.

*Remark 8* (Sign, boundedness, and shorting). The sign of  $F - V_0$  is not fixed: symmetric noise can reduce expected out-of-sample variance slightly at moderate  $\gamma$ . Nor does the noise cost blow up while Assumption 1 holds: the weights are a continuous function on the compact set  $K$ , hence bounded, say  $\|w\| \leq L$ , and  $w^\top \Sigma w \leq \lambda_{\max}(\Sigma)L^2$ . A tempting sharper bound fails. The weights sum to one but are not always long-only, so  $w^\top \Sigma w \leq \lambda_{\max}(\Sigma)$  is not available. The solved family makes this exact: at  $e = c$  on the upper noise branch,  $\gamma = 1$  gives the group weight  $x_1(2c) = \frac{4}{3}$  and asset weights  $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{6}, -\frac{1}{6})$ . The estimated covariance there is still positive definite; shorting the volatile block is simply what minimum variance does once the estimated cross covariance exceeds the smaller block variance. What breaks at large noise is also worth stating precisely. The allocation is a rational function of its inputs and becomes undefined only where a denominator vanishes, as at the point  $\gamma z = a$  of Lemma 3. Leaving the positive-definite cone does not by itself undefine the recursion; it removes the variance interpretation of the fitness, and implementations must repair or refuse.

## 8 Closing

Under cross-block estimation error the Schur coupling is a bias–variance dial, and the three layers say when to turn it. Small noise moves the optimum uniformly away from hierarchical risk

parity whenever coupling has population value (Theorem 1), at a cost of order  $\gamma\tau^2$  (Theorem 2). At an exact minimum-variance endpoint, positive marginal noise sensitivity pushes the optimum inside by order  $\tau^2$  (Theorem 3). And in the solved family the exact optimizer is unique, interior, and slides monotonically from full coupling at zero noise to the middle of the bridge at the largest admissible noise (Theorem 4). The exact examples mark what cannot be claimed in general: hierarchical risk parity can become locally optimal at a finite noise level (Example 1), and the surrogate frontier is a clipped inverse, with each endpoint owning an interval of noise levels unless its limit condition holds (Theorem 5). A walk-forward bridge on Dow constituents is interactive at [schur.microprediction.org](https://schur.microprediction.org); its minimum sits near  $\gamma \approx 0.55$  and survives Ledoit–Wolf shrinkage of the inputs. Every theorem in this paper has a live check at [schur.microprediction.org/theorems.html](https://schur.microprediction.org/theorems.html), computed from the same recursion. Neither school dominates the other. Each may own an interval of noise levels, and in the second-order surrogate the border between them is drawn at  $\tau^2 = -V'_0(\gamma)/G'(\gamma)$ .

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